

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1906

Lone Tree 138 kV
November 23, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1906. The Interconnection Customer proposes a 406.5 MW solar photovoltaic facility interconnecting at the Lone Tree 138 kV substation in the Virginia area. The request was received on October 12, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 644 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$62,682,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Michigan Power Lp - Osceola 138.0 kV is overloaded to 197.0% of its 571 MVA emergency rating for the loss of the Craven County Wood Energy Lp - Lone Tree 138.0 kV. The project contributes 100.0% DFAX (406.4 MW) to this overload.

For the loss of Craven County Wood Energy Lp Substation - Lone Tree 138.0 kV, the Lone Tree - Osceola 138.0 kV loads to 195.9% of its 578 MVA rating. Project DFAX: 100.0%.

For the loss of Asheville - Ladysmith Hydro 138.0 kV Line 1, the Ladysmith Hydro - Asheville Substation 138.0 kV loads to 186.8% of its 336 MVA rating. Project DFAX: 5.5%.

Fudge Hollow - Imhoff Tap 1 138.0 kV L: 181.2% loading (183 MVA) following the outage of the Lowmoor 345 kV/138 kV Autotransformer; DFAX = 16.8%.

For the loss of Osceola Substation - Lone Tree 138.0 kV L, the Craven County Wood Energy Lp - Lowmoor 138.0 kV loads to 178.0% of its 633 MVA rating. Project DFAX: 100.0%.

For the loss of Osceola Substation - Lone Tree Substation 138.0 kV, the Craven County Wood Energy Lp - Lone Tree 138.0 kV loads to 177.2% of its 639 MVA rating. Project DFAX: 100.0%.

The Lowmoor Substation - Fudge Hollow 138.0 kV is overloaded to 174.1% of its 190 MVA emergency rating for the loss of the Lowmoor Substation 345/138kV XFMR. The project contributes 16.8% DFAX (68.3 MW) to this overload.

Imhoff Tap 1 - Ola 138.0 kV: 172.8% loading (181 MVA) following the outage of the Lowmoor 345/138kV Autotransformer; DFAX = 16.8%.

For the loss of Asheville - Ladysmith Hydro 138.0 kV, the Asheville - Ladysmith Hydro 138.0 kV Ckt 1 loads to 167.4% of its 375 MVA rating. Project DFAX: 5.5%.

The Ola - Crockett Reit 138.0 kV is overloaded to 146.7% of its 201 MVA emergency rating for the loss of the Lowmoor 345/138kV Transformer. The project contributes 16.8% DFAX (68.3 MW) to this overload.

The Neely - Blue Ridge 138.0 kV is overloaded to 144.7% of its 358 MVA emergency rating for the loss of the Craven County Wood Energy Lp Substation - Lone Tree 138.0 kV line. The project contributes 42.8% DFAX (174.2 MW) to this overload.

For the loss of Osceola - Lone Tree Substation 138.0 kV, the Lowmoor 345-138 kV Transformer Bank loads to 140.8% of its 586 MVA rating. Project DFAX: 84.0%.

For the loss of Lowmoor 345-138 kV Auto, the Crockett Reit - Great Falls 138.0 kV loads to 138.1% of its 210 MVA rating. Project DFAX: 16.8%.

Michigan Power Lp - Blue Ridge 138.0 kV line: 137.9% loading (422 MVA) following the outage of the Craven County Wood Energy Lp - Lone Tree 138.0 kV L; DFAX = 57.9%.

For the loss of Craven County Wood Energy Lp - Lone Tree 138.0 kV, the Michigan Power Lp - Skiffes Crk 138.0 kV loads to 133.8% of its 398 MVA rating. Project DFAX: 42.0%.

For the loss of Craven County Wood Energy Lp - Lone Tree 138.0 kV L, the Vc Porterdales Hydroelectric - Neely 138.0 kV line loads to 128.0% of its 390 MVA rating. Project DFAX: 42.8%.

Huntington Court - Bonsack 138.0 kV: 127.8% loading (352 MVA) following the outage of the Lone Tree - Craven County Wood Energy Lp 138.0 kV; DFAX = 30.6%.

The Berlin Gorham - Skiffes Crk 138.0 kV is overloaded to 126.0% of its 412 MVA emergency rating for the loss of the Craven County Wood Energy Lp - Lone Tree 138.0 kV. The project contributes 42.0% DFAX (170.9 MW) to this overload.

The Bonsack - Vc Porterdales Hydroelectric 138.0 kV is overloaded to 125.4% of its 386 MVA emergency rating for the loss of the Lone Tree - Craven County Wood Energy Lp 138.0 kV. The project contributes 30.6% DFAX (124.6 MW) to this overload.

For the loss of Lowmoor 345/138 kV Transformer Bank, the Superior - Great Falls 138.0 kV loads to 123.9% of its 220 MVA rating. Project DFAX: 16.8%.

For the loss of Lowmoor 345/138kV Autotransformer, the Superior - Elmwest 138.0 kV loads to 120.3% of its 223 MVA rating. Project DFAX: 16.8%.

For the loss of Lone Tree - Craven County Wood Energy Lp 138.0 kV Line, the Florida - Berlin Gorham 138.0 kV loads to 120.1% of its 423 MVA rating. Project DFAX: 42.0%.

For the loss of Lone Tree - Craven County Wood Energy Lp 138.0 kV line, the Florida - N Huntington 138.0 kV loads to 117.2% of its 428 MVA rating. Project DFAX: 42.0%.

City W Diesel Plant - Roberts Sw Sta 345.0 kV Line: 110.0% loading (1,471 MVA) following the outage of the Navoo - Roberts Sw Sta 345.0 kV; DFAX = 5.1%.

Ashley Substation - Sherburne 138.0 kV: 107.5% loading (341 MVA) following the outage of the Bath County - Sherburne Substation 345.0 kV; DFAX = 6.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Michigan Power Lp - Osceola 138.0 kV	Rebuild line with higher-capacity conductor	\$11,675,000	42% (\$4,933,000)
Lone Tree - Osceola 138.0 kV	Rebuild line with higher-capacity conductor	\$4,475,000	42% (\$1,891,000)
Ladysmith Hydro - Asheville Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$5,700,000	14% (\$781,000)
Imhoff Tap 1 Substation - Fudge Hollow 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,038,000	36% (\$2,150,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lowmoor - Craven County Wood Energy Lp 138.0 kV	Rebuild line with higher-capacity conductor	\$10,996,000	42% (\$4,646,000)
Craven County Wood Energy Lp - Lone Tree 138.0 kV	Rebuild line with higher-capacity conductor	\$15,281,000	42% (\$6,457,000)
Fudge Hollow - Lowmoor 138.0 kV L	Rebuild line with higher-capacity conductor	\$9,072,000	36% (\$3,230,000)
Imhoff Tap 1 - Ola 138.0 kV	Rebuild line with higher-capacity conductor	\$7,538,000	36% (\$2,684,000)
Asheville - Ladysmith Hydro 138.0 kV Ckt 1	Rebuild line with higher-capacity conductor	\$6,541,000	14% (\$897,000)
Ola - Crockett Reit 138.0 kV	Rebuild line with higher-capacity conductor	\$5,759,000	36% (\$2,051,000)
Neely - Blue Ridge 138.0 kV	Rebuild line with higher-capacity conductor	\$4,463,000	31% (\$1,382,000)
Lowmoor 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$28,290,000	39% (\$11,068,000)
Crockett Reit - Great Falls 138.0 kV	Rebuild line with higher-capacity conductor	\$897,000	36% (\$319,000)
Micigan Power Lp - Blue Ridge 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,205,000	33% (\$1,388,000)
Michigan Power Lp - Skiffes Creek 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,705,000	30% (\$825,000)
Neely - Vc Porterdale Hydroelectric 138.0 kV	Rebuild line with higher-capacity conductor	\$2,646,000	31% (\$819,000)
Huntington Court - Bonsack Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$3,582,000	29% (\$1,033,000)
Berlin Gorham - Skiffes Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$12,899,000	30% (\$3,934,000)
Bonsack - Vc Porterdale Hydroelectric 138.0 kV	Rebuild line with higher-capacity conductor	\$3,619,000	29% (\$1,043,000)
Great Falls - Superior 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,410,000	36% (\$2,283,000)
Superior - Elmwest 138.0 kV	Rebuild line with higher-capacity conductor	\$13,383,000	36% (\$4,765,000)
Florida - Berlin Gorham 138.0 kV	Rebuild line with higher-capacity conductor	\$3,365,000	30% (\$1,026,000)
Florida - N Huntington 138.0 kV	Rebuild line with higher-capacity conductor	\$8,429,000	30% (\$2,571,000)
Roberts Sw Sta - City W Diesel Plant 345.0 kV Line	Reconductor line	\$2,899,000	3% (\$81,000)
Ashley Substation - Sherburne 138.0 kV	Reconductor line	\$1,695,000	25% (\$425,000)
	Total allocated to project		\$62,682,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.